

Marco Mo

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SUMMARY

Data Scientist spanning predictive modeling, NLP, time-series analysis and process mining across fraud, research and production. Delivers reproducible pipelines, validated models and dashboards, translating analytical results for directors, operators and students.

EXPERIENCE

Process Optimization, Automation & AI Specialist Aug 2026 – Present
Multi-Post

- Consolidated twelve preprocessing, reporting and audit workflows into a staff-used application.
- Reduced selected artwork preparation from approximately 30 minutes or more to seconds.
- Built validated production and machine-event dashboards from department-manager requirements.

Junior AI Specialist Jun 2026 – Aug 2026
ADC Nederland

- Developed four systems for order ingestion, database access, agentic chat and process mining.
- Combined SQL analytics, typed APIs, privacy controls, human review and automated testing.

Teaching Assistant Sep 2021 – Jul 2025
Eindhoven University of Technology

- Guided 80+ students weekly in Python, SQL, machine learning and analytical reasoning.
- Diagnosed methodological errors; coached teams on scoping, interpretation and communication.

PROJECTS

Fraud Detection MLOps Pipeline — IEEE-CIS

- Built preprocessing workflows for approximately 600,000 transactions and 400+ features.
- Prevented temporal leakage through chronological splits and held-out streaming evaluation.
- Achieved 0.9287 ROC-AUC and 0.5960 average precision with XGBoost.
- Served models through FastAPI; tracked experiments in MLflow and monitored feature drift.

Reward-Free Safe Reinforcement Learning — MSc Thesis

- Extended CEM and RENYI exploration with safety constraints and alternative entropy measures.
- Implemented PyTorch critics, PPO-style optimization and VAE density estimation.
- Evaluated coverage and safety across three environments using 24 fixed seeds.
- Found CEM with Shannon entropy achieved highest feasible coverage in classic-control tasks.

Health Platform Text Classification — Bachelor End Project

- Analyzed text from 5,087 health companies using a reproducible coding scheme.
- Compared classifiers using text representations, SMOTE, cross-validation and independent test sets.
- Expanded labels through active learning; achieved 0.610 test F1 with SVM.

EDUCATION

MSc Data Science and Artificial Intelligence 2023 – 2025
Eindhoven University of Technology

- Thesis: Reward-Free Safe Reinforcement Learning Exploration Using Different Entropy Measures

BSc Data Science 2020 – 2023
Eindhoven University of Technology and Tilburg University

- Bachelor End Project: Health Platform Text Classification Using Active Learning

SKILLS

Programming: Python, SQL, R, C++

Machine Learning: scikit-learn, XGBoost, PyTorch, TensorFlow, NLP

Methods: cross-validation, forecasting, active learning, class imbalance, safe reinforcement learning

Data and MLOps: ETL, schema validation, MLflow, Docker, CI/CD

Analytics: process mining, pandas, NumPy, Dash, Plotly

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)